

Android SDK Manual

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1. Overview

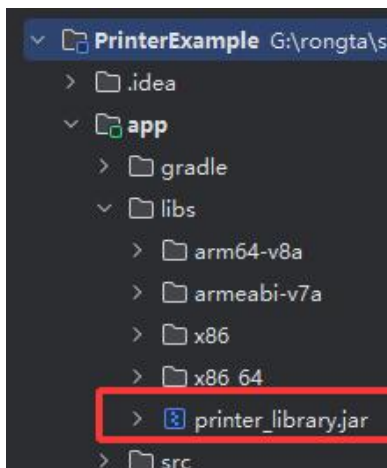
printer_library is a jar library for controlling printers under the Android platform, which facilitates the interaction between APP and printer information. The SDK mainly includes the connection of the printer, and the connection interface is Bluetooth, USB, WIFI and serial port. It also contains interfaces for printing instructions, for example
Print text, bar code, pictures, etc

2. Integration Steps

By integrating the Android SDK into the application, the default developer is already familiar with the basic use of the IDE (Eclipse or Android Studio) and has a certain knowledge base of Android programming. Development environment: Java 8, Android Studio 3.2 or above;

2.1 find app > libs, if there is no libs, create a new one; If the project is exported from AS, the libs directory is directly in the root directory, remember this;

2.2 printer_library.jar jarCopy the package to the libs package, as shown below:



2.3 Find build.gradle(Module:app), then find dependencies and add :implementation files('libs/printer_library.jar') as shown below:

```
dependencies {
    implementation fileTree(include: ['*.jar'], dir: 'libs')
    implementation 'androidx.constraintlayout:constraintlayout:1.1.3'
    implementation 'androidx.annotation:annotation:1.4.0'
    testImplementation 'junit:junit:4.13.2'
    implementation 'androidx.appcompat:appcompat:1.4.1'
```

3. Permissions Request

AndroidManifest.xml

```
<uses-feature android:name="android.hardware.usb.host" />
<uses-permission android:name="android.permission.BLUETOOTH_SCAN" />
<uses-permission android:name="android.permission.BLUETOOTH_ADVERTISE" />
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.CHANGE_NETWORK_STATE" />
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE" />
<uses-permission android:name="android.permission.WAKE_LOCK" />
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.READ_MEDIA_IMAGES" tools:targetApi="33" />
```

MainActivity


```

private String[] NEED_PERMISSION;
    private List<String> NO_PERMISSION = new ArrayList<String>();
private void CheckAllPermission() {
    NO_PERMISSION.clear();
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        NEED_PERMISSION = new String[]{Manifest.permission.READ_MEDIA_IMAGES,
            Manifest.permission.BLUETOOTH_CONNECT,
            Manifest.permission.BLUETOOTH_SCAN,
            Manifest.permission.BLUETOOTH_ADVERTISE,
            Manifest.permission.ACCESS_FINE_LOCATION
        };
    }else if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
        NEED_PERMISSION = new
String[]{Manifest.permission.READ_EXTERNAL_STORAGE,
            Manifest.permission.BLUETOOTH_CONNECT,
            Manifest.permission.BLUETOOTH_SCAN,
            Manifest.permission.BLUETOOTH_ADVERTISE,
            Manifest.permission.ACCESS_FINE_LOCATION
        };
    } else {
        NEED_PERMISSION = new
String[]{Manifest.permission.READ_EXTERNAL_STORAGE,
            Manifest.permission.ACCESS_FINE_LOCATION,
            Manifest.permission.BLUETOOTH_ADMIN
        };
    }
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.M) {
        for (int i = 0; i < NEED_PERMISSION.length; i++) {
            if (checkSelfPermission(NEED_PERMISSION[i]) !=
PackageManager.PERMISSION_GRANTED) {
                NO_PERMISSION.add(NEED_PERMISSION[i]);
            }
        }
        if (NO_PERMISSION.size() == 0) {
            recordVideo();
        } else {
            requestPermissions(NO_PERMISSION.toArray(new
String[NO_PERMISSION.size()]), REQUEST_CAMERA);
        }
    } else {
        recordVideo();
    }
}
}

```

4. Connect to printer

Printer connection methods: WiFi, USB, Com, connection steps are as follows:

4.1 Initialize Printer basic class

```
private RTPrinter rtPrinter = null;
private PrinterFactory printerFactory;
private Object configObj;
private ArrayList<PrinterInterface> printerInterfaceArrayList = new ArrayList<>();
private PrinterInterface curPrinterInterface = null;
public void init() {
    printerFactory = new UniversalPrinterFactory();
    rtPrinter = printerFactory.create();
    PrinterObserverManager.getInstance().add(this);//Add connection status listener
}
```

4.1.1 Printer connection status monitoring

```
public void printerObserverCallback(final PrinterInterface printerInterface, final int state)
```

Parameters:

printerInterface -PrinterInterface

State - CommonEnum.CONNECT_STATE_INTERRUPTED : connected disconnected

- CommonEnum.CONNECT_STATE_SUCCESS :connected successful

Example

```
@Override
public void printerObserverCallback(final PrinterInterface printerInterface, final int state) {
    runOnUiThread(new Runnable() {
        @Override
        public void run() {
            switch (state) {
                case CommonEnum.CONNECT_STATE_SUCCESS:
                    curPrinterInterface = printerInterface;
                    printerInterfaceArrayList.add(printerInterface);
                    rtPrinter.setPrinterInterface(printerInterface);
                    BaseApplication.getInstance().setRtPrinter(rtPrinter);
                    setPrintEnable(true);
                    isPrinterConnect=true;
                    break;
                case CommonEnum.CONNECT_STATE_INTERRUPTED:
                    curPrinterInterface = null;
                    printerInterfaceArrayList.remove(printerInterface);
                    BaseApplication.getInstance().setRtPrinter(null);
                    setPrintEnable(false);
                    isPrinterConnect=false;
                    break;
            }
        }
    })
}
```

```

    }
    });
}

```

4.1.2 Printer callback message monitoring

```

public void printerReadMsgCallback(PrinterInterface printerInterface, final byte[]
bytes)

```

Parameters:

printerInterface -Current printer interface
bytes - printer callback message

Example:

```

/**
 * 1 print status
 * 2 cash box
 * 3 Firmware Version
 */
private int msgType=1;
@Override
public void printerReadMsgCallback(PrinterInterface printerInterface, final byte[] bytes) {
    Log.i(TAG, "printerReadMsgCallback: "+FuncUtils.ByteArrToHex(bytes));
    String msg;
    switch (msgType){
        case 1://Accept printer status here
            PrinterStatusBean StatusBean = PrinterStatusPareseUtils.parsePrinterStatusResult(bytes);
            msg = PrinterStatusPareseUtils.getPrinterStatusStr(StatusBean);
            int statusCode = PrinterStatusPareseUtils.getPrinterStatusInt(StatusBean);
            Log.i(TAG, "onPrinterStatus: " +statusCode+" "+msg);
            toastUI("print status: " +statusCode+" "+msg);
            break;
        case 2://cash box status
            int scashbox = PrinterStatusPareseUtils.parseCashdrawerStatus(bytes);
            Log.i(TAG, "cash box status: " + scashbox);
            if (scashbox == 1) {
                toastUI("Cashdrawer Open");
            } else if (scashbox == 0) {
                toastUI("Cashdrawer Close");
            }
            break;
        case 3://Firmware version
            int msgLen=bytes.length;
            if(BaseApplication.getInstance().getCurrentCmdType()!=BaseEnum.CMD_ESC){
                if (msgLen > 6) {
                    int l = msgLen-6;
                    byte[] subByteArr = subByteArr(bytes, 6, l);
                    msg= getByteArr2Str(subByteArr);
                }
            }
    }
}

```

```

        toastUI(msg);
    }
} else {
    if (msgLen > 3) {
        msg= FuncUtils.getByteArr2Str(bytes);
        toastUI(msg);
    }
}
break;
}
}

```

4.2 Connection

public void connectDevice(Object device)

Printer API connection status will be IPrintMsgCallbackListener feedback

Parameters:

```

device    --WiFiConfigBean
          --UsbConfigBean
          --BluetoothEdrConfigBean
          --SerialPortConfigBean

```

4.2.1WiFi Connection

UDP Scan for IP, new IpNetSearch().start()

Example:

```

new IpNetSearch() {
    @Override
    public void onSearchStart() {
        Message message = new Message();
        message.what = SCAN_START;
        handler.sendMessage(message);
    }

    @Override
    public void onSearchFinish(List devicesList) {
        Message message = new Message();
        message.obj = devicesList;
        message.what = SCAN_FINISH;
        handler.sendMessage(message);
    }

    @Override
    public void onSearchError(String msg) {
        Message message = new Message();
        message.what = SCAN_ERROR;
    }
}

```

```

        message.obj = msg;
        handler.sendMessage(message);
    }
}.start();
private List<IPDeviceBean> deviceBeanList;
//IP sacn add
private final static int SCAN_START = 1001;
private final static int SCAN_FINISH = 1002;
private final static int SCAN_ERROR = 1003;
private final static int SHOW_MSG = 1004;
@SuppressWarnings("HandlerLeak")
private Handler handler = new Handler() {
    @Override
    public void handleMessage(Message msg) {
        switch (msg.what) {
            case SHOW_MSG:
                ToastUtil.show(getApplicationContext(), (String) msg.obj);
                break;
            case SCAN_START:
                Log.d(TAG, "SCAN_START-----");
                break;
            case SCAN_FINISH:
                deviceBeanList = (List<IPDeviceBean>) msg.obj;
                if(deviceBeanList.size()>0)
                    Log.d(TAG, "SCAN_FINISH-----"+deviceBeanList.get(0).toString());
                pb_connect.setVisibility(View.GONE);
                showIpDialog(getString(R.string.tip_select_ip_device));
                break;
            case SCAN_ERROR:
                pb_connect.setVisibility(View.GONE);
                showIpDialog(getString(R.string.tip_select_ip_device));
                break;
        }
    }
};

```

Connect to the printer via a WificonfigBean over WiFi

Example:

```

WiFiConfigBean wiFiConfigBean = (WiFiConfigBean) configObj;
PIFactory piFactory = new WiFiFactory();
PrinterInterface printerInterface = piFactory.create();
printerInterface.setConfigObject(wiFiConfigBean);
rtPrinter.setPrinterInterface(printerInterface);
try {
    rtPrinter.connect(wiFiConfigBean);
}

```

```
    } catch (Exception e) {  
        e.printStackTrace();  
    }  
}
```

4.2.2 USB Connection

Get USB device

Example:

```
public class UsbDeviceChooseDialog extends DialogFragment {  
    private final String TAG = getClass().getSimpleName()+"ComPrinter";  
    private Context mContext;  
    private ListView lvContent;  
    private TextView tvEmpty;  
    private AdapterView.OnItemClickListener mListener;  
    private List<UsbDevice> mList;  
    private UsbDeviceAdapter mAdapter;  
    private UsbManager mUsbManager;  
    private UsbDeviceReceiver mUsbReceiver;  
    @Override  
    public void onAttach(Activity activity) {  
        super.onAttach(activity);  
        mContext = activity;  
    }  
    @NonNull  
    @Override  
    public Dialog onCreateDialog(Bundle savedInstanceState) {  
        LayoutInflater inflater = getActivity().getLayoutInflater();  
        View view = inflater.inflate(R.layout.dialog_choose_usb_device, null);  
        initView(view);  
        setListener();  
        initData();  
        setAdapter();  
        registerUsbReceiver();  
        AlertDialog.Builder builder = new AlertDialog.Builder(mContext);  
        builder.setView(view).setCancelable(true).setNegativeButton(R.string.dialog_cancel, null);  
        return builder.create();  
    }  
    private void registerUsbReceiver() {  
        Log.i(TAG, "registerUsbReceiver: ");  
        mUsbReceiver = new UsbDeviceReceiver(new UsbDeviceReceiver.CallBack() {  
            @Override  
            public void onPermissionGranted(UsbDevice usbDevice) {  
                LogUtils.d(TAG,"onPermissionGranted:"+usbDevice.getDeviceId());  
            }  
            @Override  
            public void onDeviceAttached(UsbDevice usbDevice) {
```

```

        LogUtils.d(TAG,"onDeviceAttached:"+usbDevice.getDeviceId());
        mList.add(usbDevice);
        mAdapter.notifyDataSetChanged();
        tvEmpty.setVisibility(View.GONE);
    }

    @Override
    public void onDeviceDetached(UsbDevice usbDevice) {
        LogUtils.d(TAG,"onDeviceDetached:"+usbDevice.getDeviceId());
        mList.remove(usbDevice);
        mAdapter.notifyDataSetChanged();
        if (mList.size() == 0) {
            tvEmpty.setVisibility(View.VISIBLE);
        }
    }

});

IntentFilter intentFilter = new IntentFilter();
intentFilter.addAction(UsbManager.ACTION_USB_DEVICE_ATTACHED);
intentFilter.addAction(UsbManager.ACTION_USB_DEVICE_DETACHED);
mContext.registerReceiver(mUsbReceiver, intentFilter);
}

private void initView(View view) {
    lvContent = (ListView) view.findViewById(R.id.lv_dialog_choose_usb_device);
    tvEmpty = (TextView) view.findViewById(R.id.tv_dialog_choose_usb_device_empty);
}

private void setListener() {
    lvContent.setOnItemClickListener(mListener);
}

private void initData() {
    mList = new ArrayList<>();
    mUsbManager = (UsbManager) mContext.getSystemService(Context.USB_SERVICE);
    HashMap<String, UsbDevice> deviceList = mUsbManager.getDeviceList();
    LogUtils.d(TAG, "deviceList size = " + deviceList.size());
    Iterator<UsbDevice> deviceIterator = deviceList.values().iterator();
    while (deviceIterator.hasNext()) {
        UsbDevice device = deviceIterator.next();
        LogUtils.d(TAG, "device getDeviceName" + device.getDeviceName());
        LogUtils.d(TAG, "device getVendorId" + device.getVendorId());
        LogUtils.d(TAG, "device getProductId" + device.getProductId());
        mList.add(device);
    }
    if (mList.size() == 0) {
        tvEmpty.setVisibility(View.VISIBLE);
    }
}
}

```

```

private void setAdapter() {
    mAdapter = new UsbDeviceAdapter(mContext, mList);
    lvContent.setAdapter(mAdapter);
}

public void setOnItemClickListener(AdapterView.OnItemClickListener listener) {
    mListener = listener;
}

@Override
public void onDismiss(DialogInterface dialog) {
    super.onDismiss(dialog);
    LogUtils.v(TAG, "onDismiss");
    mContext.unregisterReceiver(mUsbReceiver);
}
}

private void showUSBDeviceChooseDialog() {
    final UsbDeviceChooseDialog usbDeviceChooseDialog = new UsbDeviceChooseDialog();
    usbDeviceChooseDialog.setOnItemClickListener(new AdapterView.OnItemClickListener() {
        @Override
        public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
            UsbDevice mUsbDevice = (UsbDevice) parent.getAdapter().getItem(position);
            PendingIntent mPermissionIntent = PendingIntent.getBroadcast(MainActivity.this, 0,
                new Intent(MainActivity.this.getApplicationInfo().packageName),
                PendingIntent.FLAG_IMMUTABLE);
            tv_device_selected.setText(getString(R.string.adapter_usbdevice) +
                mUsbDevice.getDeviceId()); //+ (position + 1));
            configObj = new UsbConfigBean(BaseApplication.getInstance(), mUsbDevice,
                mPermissionIntent);
            tv_device_selected.setTag(BaseEnum.HAS_DEVICE);
            isConfigPrintEnable(configObj);
            usbDeviceChooseDialog.dismiss();
        }
    });
    usbDeviceChooseDialog.show(getFragmentManager(), null);
}

```

Connect to the printer by device

Example:

```

UsbConfigBean usbConfigBean = (UsbConfigBean) configObj;
UsbManager mUsbManager = (UsbManager) getSystemService(Context.USB_SERVICE);
PIFactory piFactory = new UsbFactory();
PrinterInterface printerInterface = piFactory.create();
printerInterface.setConfigObject(usbConfigBean);
rtPrinter.setPrinterInterface(printerInterface);
if (mUsbManager.hasPermission(usbConfigBean.usbDevice)) {
    try {

```



```

        rtPrinter.connect(usbConfigBean);
    } catch (Exception e) {
        e.printStackTrace();
    }
} else {
    mUsbManager.requestPermission(usbConfigBean.usbDevice, usbConfigBean.pendingIntent);
}

```

4.2.3 Bluetooth Connection

Get Bluetooth Device

Example:

```

private void showBluetoothDeviceChooseDialog() {
    BluetoothDeviceChooseDialog bluetoothDeviceChooseDialog = new BluetoothDeviceChooseDialog();
    bluetoothDeviceChooseDialog.setOnDeviceItemClickListener(new
BluetoothDeviceChooseDialog.onDeviceItemClickListener() {
        @Override
        public void onDeviceItemClick(BluetoothDevice device) {
            if (TextUtils.isEmpty(device.getName())) {
                tv_device_selected.setText(device.getAddress());
            } else {
                tv_device_selected.setText(device.getName() + " [" + device.getAddress() + "]");
            }
            configObj = new BluetoothEdrConfigBean(device);
            tv_device_selected.setTag(BaseEnum.HAS_DEVICE);
            isConfigPrintEnable(configObj);
        }
    });
    bluetoothDeviceChooseDialog.show(MainActivity.this.getFragmentManager(), null);
}

```

Connect to the printer by device

Example:

```

BluetoothEdrConfigBean bluetoothEdrConfigBean = (BluetoothEdrConfigBean) configObj;
PIFactory piFactory = new BluetoothFactory();
PrinterInterface printerInterface = piFactory.create();
printerInterface.setConfigObject(bluetoothEdrConfigBean);
rtPrinter.setPrinterInterface(printerInterface);
try {
    rtPrinter.connect(bluetoothEdrConfigBean);
} catch (Exception e) {
    e.printStackTrace();
}

```

4.2.4 Com Connection

Get Com Device

Example:

```

/**

```

```

    * init com device
    */
private void initComDevice() {
    SerialPortFinder serialPortFinder = new SerialPortFinder();
    // device
    mDevices = serialPortFinder.getAllDevicesPath2();
    if (mDevices.length == 0) {
        mDevices = new String[]{
            getString(R.string.str_no_devices)
        };
    }
    // Baud rate
    mBaudrates = getResources().getStringArray(R.array.baudrates);
    mDeviceIndex = mDevices.length - 1;
    mDeviceIndex = mDeviceIndex >= mDevices.length ? mDevices.length - 1 : mDeviceIndex;
    mBaudrateIndex = 0;
}

private void initSpinners(){
    ArrayAdapter<String> deviceAdapter =
        new ArrayAdapter<String>(this,R.layout.spinner_default_item,mDevices);
    deviceAdapter.setDropDownViewResource(R.layout.spinner_item);
    mSpComPath.setAdapter(deviceAdapter);
    mSpComPath.setOnItemSelectedListener(this);

    ArrayAdapter<String> baudrateAdapter =
        new ArrayAdapter<String>(this, R.layout.spinner_default_item, mBaudrates);
    baudrateAdapter.setDropDownViewResource(R.layout.spinner_item);
    mSpBps.setAdapter(baudrateAdapter);
    mSpBps.setOnItemSelectedListener(this);

    mSpComPath.setSelection(mDeviceIndex);
    mSpBps.setSelection(mBaudrateIndex);
}

/**
 * com
 */
private void showCOMDeviceChooseDialog() {
    View comSelectView = View.inflate(this,R.layout.dialog_com_select,null);
    TextView title = comSelectView.findViewById(R.id.tv_dialog_title);
    title.setText("Title");
    mSpComPath = comSelectView.findViewById(R.id.sp_com_path);
}

```

```

mSpBps = comSelectView.findViewById(R.id.sp_bps);
initComDevice();
initSpinners();
Button btn_cancle = comSelectView.findViewById(R.id.negativeButton);
Button btn_confirm = comSelectView.findViewById(R.id.positiveButton);
final AlertDialog.Builder builder = new AlertDialog.Builder(this);
builder.setView(comSelectView);
final AlertDialog dialog = builder.create();
dialog.setCanceledOnTouchOutside(false);
dialog.show();

btn_cancle.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        dialog.dismiss();
    }
});

btn_confirm.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        tv_device_selected.setText(mDevices[mDeviceIndex]+" - "+mBaudrates[mBaudrateIndex]);
        tv_device_selected.setTag(BaseEnum.HAS_DEVICE);
        dialog.dismiss();
    }
});
}

@Override
public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {
    switch (adapterView.getId()) {
        case R.id.sp_com_path:
            mDeviceIndex = i;
            mSerialPortConfigBean.setFile(new File(mDevices[mDeviceIndex]));
            break;
        case R.id.sp_bps:
            mBaudrateIndex = i;
            mSerialPortConfigBean.setBaudrate(Integer.parseInt(mBaudrates[mBaudrateIndex]));
            break;
    }
}

@Override
public void onNothingSelected(AdapterView<?> adapterView) {
}

```

Connect to the printer by device

Example:

```
SerialPortConfigBean mSerialPortConfigBean = new SerialPortConfigBean();
PIFactory piFactory = new SerialPortFactory();

PrinterInterface printerInterface = piFactory.create();
printerInterface.setConfigObject(mSerialPortConfigBean);
rtPrinter.setPrinterInterface(printerInterface);

try {
    rtPrinter.connect(serialPortConfigBean);
} catch (Exception e) {
    e.printStackTrace();
}
```

4.3 Disconnect

```
public void disconnect()
```

Example:

```
private void disconnectPrinter(){
    if (rtPrinter != null && rtPrinter.getPrinterInterface() != null) {
        rtPrinter.disconnect();
    }
}
```

5. RTPrinter

Operation: connect, disconnect, write messages, read message, get the current printer interface(connection object).

5.1 connect the printer

```
connect(T object)
```

Parameters:

```
Object -object (
    WiFi:WiFiConfigBean,
    Bluetooth: BluetoothEdrConfigBean,
    USB: UsbConfigBean)
```

5.2 write data

```
writeMsg(byte[] bytes)
```

write data

Parameters:

bytes -

5.3 write data Async

```
writeMsgAsync( byte[] bytes)
```

write data

Parameters:

bytes -

5.4 Read the printer's msgs

```
readMsg()
```

Parameters:

Return:

bytes -printer's msgs which is sent back

5.5 Get the printer interface

getPrinterInterface()

Parameters:

Return: printer interface

5.6 Set printer interface

setPrinterInterface(PrinterInterface printerInterface)

Parameters:

printerInterface - PrinterInterface

Return:

6. CommonSetting

6.1 public void setAbsolutionPositionN(int absolutionPositionN)

- Suitable for Esc, pin printing

Absolute fixed point positioning

Based on the current left limit, shift the print head right $(n1+n2 \times 256) / 60$ inches. If the distance exceeds the right-hand limit, this setting is invalid. The maximum setting distance is 10.6 inches.

Parameters:

absolutionPositionN - $n \leq 636$; $n1 = n \% 256$; $n2 = n / 256$;

6.2 public void setPageWidth(int pageWidth)

For Cpcl, ESC

page width setting

Parameters:

pageWidth - CPCL:In-Dot ESC IN-mm: n

6.3 public LableSizeBean getLableSizeBean()

Get label size object

Returns:

6.4 public void setLableSizeBean(LableSizeBean lableSizeBean)

Label size setting

-for Tsc, Zpl

-For Tsc, Zpl

Parameters:

lableSizeBean -

getLabelGap

public int getLabelGap()

Get Label Gap

Returns:

6.5 public void setAlign(int align)

For Pin, Esc

Esc(ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT)]

align setting [Esc only support ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT]

Parameters:

align - ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT, ALIGN_BOTH_SIDES

[Esc only support ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT]

6.6 public int getEscLineSpacing()

6.7 public void setEscLineSpacing(int n)

For ESC

EPOS Line space setting, $n * 0.125$ mm

Parameters:

$n - n * 0.125$ mm , range: $0 \leq n \leq 255$

6.8 public int getEscCharSpacing()

6.9 public void setEscCharSpacing(int n)

6.10 public byte[] setPointXY(int x, int y)

6.11 public byte[] setBlackLabelPlus(int plus)

6.12 public byte[] setBlackLabelSub(int sub)

6.14 public int getxSimLineSpacing()

6.15 public void setxSimLineSpacing(int spacing)

This command sets the line spacing between successive printed text lines (dot row). Each dot row is 0.125mm. $0 \leq n \leq 40$) decimal= . If n value is greater than 0x28 or Decimal 40, then n will be set to 0x28)Decimal 40).

Parameters:

spacing - 0-40

7 TextSetting

7.1 public SettingEnum getBold()

7.2 public void setBold(SettingEnum setting)

For ESC, Pin

Bold Setting D

Parameters:

setting - Enable, Disable

public SettingEnum getItalic()

public void setItalic(SettingEnum setting)

italic setting

Parameters:

setting - Enable, Disable

7.3 public SettingEnum getUnderline()

7.4 public void setUnderline(SettingEnum underline)

For ESC, Pin

underline setting

Parameters:

underline - Enable, Disable

7.5 public SettingEnum getDoublePrinting()

7.6 public void setDoublePrinting(SettingEnum doublePrinting)

For Pin

double printint setting

Parameters:

doublePrinting - Enable, Disable

7.7 public int getAlign()

7.8 public void setAlign(int align)

For Pin, ESC

ESC(ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT)]

align setting [ESC only support ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT]

Parameters:

align - ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT, ALIGN_BOTH_SIDES

[ESC only support ALIGN_LEFT, ALIGN_MIDDLE, ALIGN_RIGHT]

7.9 public SettingEnum getDoubleWidth()

7.10 public void setDoubleWidth(SettingEnum doubleWidth)

For ESC, Pin

double width setting

Parameters:

doubleWidth - Enable, Disable

7.11 public SettingEnum getDoubleHeight()

7.12 public void setDoubleHeight(SettingEnum doubleHeight)

For ESC, Pin

double height setting

Parameters:

doubleHeight - Enable, Disable

7.13 public int getFontStyle()

7.14 public void setFontStyle(int fontStyle)

For Pin

Font Style setting

Parameters:

fontStyle - FONT_STYLE_NORMAL, FONT_STYLE_HOLLOW, FONT_STYLE_SHADOW,
FONT_STYLE_HOLLOW_AND_SHADOW

7.15 public int getPinPrintMode()

7.16 public void setPinPrintMode(int pinPrintMode)

For Pin

Print Mode setting

Parameters:

pinPrintMode - PIN_PRINT_MODE_Bidirectional, PIN_PRINT_MODE_ONeway,
PIN_PRINT_MODE_Full_Bidirectional

7.17 public SettingEnum getDoubleSizeChineseCharctor()

7.18 public void setDoubleSizeChineseCharctor(SettingEnum doubleSizeChineseCharctor)

Deprecated.

Deprecated, No Working

For Pin

Chinese charactor setting

Parameters:

doubleSizeChineseCharactor - accepted value:
CommonEnum.Setting.Enable/CommonEnum.Setting.Disable

7.19 public SettingEnum getIsAntiWhite()
7.20 public void setIsAntiWhite(SettingEnum isAntiWhite)
For ESC,XSim
anti white setting

Parameters:

isAntiWhite - SettingEnum.Enable = AntiWhite , SettingEnum.Disable = Cancel

7.21 public SettingEnum getIsEscSmallCharactor()
7.22 public void setIsEscSmallCharactor(SettingEnum isEscSmallCharactor)
For ESC
small charactor setting

Parameters:

isEscSmallCharactor - SettingEnum.Enable = small font,Font-B),
SettingEnum.Disable =Normal font,Font-A

7.23 public Position getTxtPrintPosition()
For Tsc, Cpcl,Zpl
Get text print position object

Returns:
Position Object

7.24 public void setTxtPrintPosition(Position txtPrintPosition)
For Tsc,Cpcl,Zpl
set text print position

Parameters:

txtPrintPosition - position , Default: x=0,y=0

7.25 public TsplFontTypeEnum getTsplFontTypeEnum()
7.26 public void setTsplFontTypeEnum(TsplFontTypeEnum tscFontTypeEnum)
For Tspl
Tspl Font Type Style

Parameters:

tscFontTypeEnum -

7.27 public int getxMultiplication()
7.28 public void setxMultiplication(int xMultiplication)
For Tspl, Cpcl, Zpl
set Font X-Multiplication

Parameters:

xMultiplication -

7.29 public int getyMultiplication()
7.30 public void setyMultiplication(int yMultiplication)
For Tspl, Cpcl, Zpl

set Font Y-Multiplication

Parameters:

yMultiplication -

7.31 public PrintRotation getPrintRotation()

7.32 public void setPrintRotation(PrintRotation printRotation)

For Tspl

Print rotation setting

For tspl text, barcode print

Parameters:

printRotation - Rotate0, Rotate90, Rotate180, Rotate270

7.33 public CpclFontTypeEnum getCpclFontTypeEnum()

7.34 public void setCpclFontTypeEnum(CpclFontTypeEnum cpclFontTypeEnum)

Cpcl Font Type Style

Parameters:

cpclFontTypeEnum -

7.35 public int getCpclFontSize()

7.36 public void setCpclFontSize(int cpclFontSize)

For Cpcl

cpcl text size setting, only support english and number

Parameters:

cpclFontSize -

7.37 public ZplFontTypeEnum getZplFontTypeEnum()

7.38 public void setZplFontTypeEnum(ZplFontTypeEnum zplFontTypeEnum)

For Zpl, Font type Setting

Parameters:

zplFontTypeEnum - 0=Vector font (downloadable font),Other resident fonts:A~G,1~9

7.39 public int getZplWidthFactor()

7.40 public void setZplWidthFactor(int zplWidthFactor)

For Zpl

width factor setting

Parameters:

zplWidthFactor -

accept value:

Vector font(FONT_DOWNLOAD_FONT): 10~32000,

Dot matrix font(Font A~G, 1~9): 1~10

7.41 public int getZplHeightFactor()

7.42 public void setZplHeightFactor(int zplHeightFactor)

For Zpl

height factor setting

Parameters:

zplHeightFactor -

accept value:

Vector font(FONT_DOWNLOAD_FONT): 10~32000,

Dot matrix font(Font A~G, 1~9): 1~10

7.43 public ESCFontTypeEnum getEscFontType()

7.44 public ESCPFontTypeEnum getEscpFontType()

7.45 public void setEscFontType(ESCFontTypeEnum escFontType)

For ESC Portable Printer,

Font Type Setting, Only support English-Number charactor

Parameters:

escFontType - Accept Value[ESCFontTypeEnum]:

FONT_A_12x24

FONT_B_9x24

FONT_C_9x17

FONT_D_8x16

7.46 public void setEscpFontType(ESCPFontTypeEnum e)

7.47 public void setPrintContent(java.lang.String e)

7.48 Public String getPrintContent()

7.49 public int getCpclTextSpacing()

7.50 public void setCpclTextSpacing(int cpclTextSpacing)

For CPCL

text spacing setting.

Parameters:

cpclTextSpacing - n >= 0

7.51 public XSimFontTypeEnum getXSimFontTypeEnum()

7.52 public void setxSimFontTypeEnum(XSimFontTypeEnum xSimFontTypeEnum)

Parameters:

xSimFontTypeEnum - FONT_0~FONT_15

8 BarcodeSetting

8.1 public BarcodeStringPosition getBarcodeStringPosition()

8.2 public BarcodeType getBarcodeType()

8.3 public void setBarcodeType(BarcodeType barcodeType)

8.4 public void setBarcodeStringPosition(BarcodeStringPosition barcodeStringPosition)

For ESC,Tsc

Set the bar code content location

Parameters:

barcodeStringPosition - NONE = [Hidden], ABOVE_BARCODE , BELOW_BARCODE,
ABOVE_BELOW_BARCODE

8.5 public EscBarcodePrintOritention getEscBarcodePrintOritention()

8.6 public String getBarcodeContent()

8.7 public void setBarcodeContent(java.lang.String b)

8.8 public int getQrcodeDotSize()

8.9 public void setQrcodeDotSize(int qrcodeDotSize)

For Esc, Tsc

QRCode dot size settings

Parameters:

qrcodeDotSize - Esc(1~15), Tsc(1~10)

8.10 public PrintRotation getPrintRotation()

8.11 public void setPrintRotation(PrintRotation printRotation)

For Tsc

barcode rotation settings

Parameters:

printRotation -

8.12 public int getHeightInDot()

8.13 public void setHeightInDot(int heightInDot)

For Tsc,Cpcl

barcode height settings

Parameters:

heightInDot - In Dot

8.14 public int getNarrowInDot()

8.15 public void setNarrowInDot(int narrowInDot)

For Tspl

narrow bar settings

Parameters:

narrowInDot - In Dot

8.16 public int getWideInDot()

8.17 public void setWideInDot(int wideInDot)

For Tspl

wide bar settings

Parameters:

wideInDot - In Dot

8.18 public Position getPosition()

8.19 public void setPosition(Position position)

For Tspl, Cpcl, Zpl

Set the bar code printing position

Parameters:

position - position.x , position.y

8.20 public QrcodeEccLevel getQrcodeEccLevel()

8.21 public void setQrcodeEccLevel(QrcodeEccLevel qrcodeEccLevel)

Error correction capability level

():L=7%,M=15%,Q=25%,H=30%

Parameters:

qrcodeEccLevel - L=7%,M=15%,Q=25%,H=30%

8.22 public int getBarcodeWidth()

8.23 public void setBarcodeWidth(int barcodeWidth)

For Esc

Bar code width, range: 2~6

barcode width, range: 2~6, default:3

Parameters:

barcodeWidth - range: 2~6, default:3

8.24 public void setEscBarcodFont(ESCBarcodeFontTypeEnum font)

8.25 public ESCBarcodeFontTypeEnum getEscBarcodFont()

8.26 public int getAZTEC_xd()

8.27 public void setAZTEC_xd(int AZTEC_xd)

8.28 public int getAZTEC_ec()

8.29 public void setAZTEC_ec(int AZTEC_ec)

8.30 public int getTsc_alignment()

8.31 public void setTsc_alignment(int tsc_alignment)

20220630 TSPL2 alignment Specify the alignment of barcode 0 : default (Left) 1 : Left 2 : Center 3 : Right

8.32 public int getHuman_readable()

8.33 public void setHuman_readable(int human_readable)

20220630 TSPL2 human readable 0: not readable 1: human readable aligns to left 2: human readable aligns to center 3: human readable aligns to right

8.34 public int getxSimPdf417EH()

8.35 public void setxSimPdf417EH(int xSimPdf417EH)

Element Height byte. This is the height length of a single element in the Pdf417 symbol, that is, bar/space height. A value between 0x03 to 0x0A can be used. Recommended Value: 0x06

Parameters:

xSimPdf417EH -

9 BitmapSetting

9.1 public void setPrintPostion(@NonNull Position printPostion)

For Tspl, Cpcl

Bitmap position setting[Position Object], Default: x=0, y=0

Parameters:

printPostion -

9.2 public BitmapLimitSizeBean getBitmapLimitSizeBean()

9.3 public void setBitmapLimitSizeBean(BitmapLimitSizeBean bitmapLimitSizeBean)

Deprecated, No work

Set the size of the picture to be printed (width, height)

Parameters:

bitmapLimitSizeBean - bitmapLimitSizeBean.limitWidth, bitmapLimitSizeBean.limitHeight

9.4 public int getBimtapLimitWidth()

9.5 public void setBimtapLimitWidth(int bimtapLimitWidth)

Bitmap Limit Width setting(N mm * 8)

Parameters:

bimtapLimitWidth -

9.6 public void setBmpPrintMode(BmpPrintMode bmpPrintMode)

For ESC, Tspl

Bitmap print mode settings

Parameters:

bmpPrintMode-BmpPrintMode.MODE_SINGLE_COLOR,
BmpPrintMode.MODE_MULTI_COLOR[Default]

10 EscCmd

10.1 public byte[] getHeaderCmd()

Description copied from class: Cmd

Get header commands

Tspl: CLS command , clear cache

Esc/Pin: printer initialization

Specified by:

getHeaderCmd in class Cmd

Returns:

10.2 public byte[] getLFCmd()

Description copied from class: Cmd

\n

Wrap

Specified by:

getLFCmd in class Cmd

Returns:

10.3 public byte[] getCRCmd()

Description copied from class: Cmd

\r

Enter

Specified by:

getCRCmd in class Cmd

Returns:

public byte[] getLFCRCmd()

Description copied from class: Cmd

Wrap + Enter

Specified by:

getLFCRCmd in class Cmd

Returns:

10.4 public byte[] getCommonSettingCmd(CommonSetting commonSetting)

Description copied from class: Cmd

Common Setting command

Specified by:

getCommonSettingCmd in class Cmd

Returns:

10.5 public byte[] getTextCmd(TextSetting textSetting,
java.lang.String text,

java.lang.String chartSetName)
throws java.io.UnsupportedEncodingException

Description copied from class: Cmd
Print text command

Specified by:

getTextCmd in class Cmd

Parameters:

textSetting - Text Setting
text - Content

chartSetName - GBK[Default]、UTF-8

Returns:

10.6 public byte[] getBitmapCmd(BitmapSetting bitmapSetting,
Bitmap bitmap)

Description copied from class: Cmd
Get Bitmap printing command

Specified by:

getBitmapCmd in class Cmd

Parameters:

bitmapSetting - Bitmap setting (width limit)
bitmap - the bitmap which to print

Returns:

byte[] cmds

10.7 public byte[] getBarcodeCmd(BarcodeType barcodeType,
BarcodeSetting barcodeSetting,
java.lang.String barcodeContent)
throws SdkException Description copied from class: Cmd
Get Barcode Printing command

Specified by:

getBarcodeCmd in class Cmd

Parameters:

barcodeType - Bar code type
barcodeSetting - Bar code setup
barcodeContent - Barcode content

Returns:

Throws:

SdkException

10.8 public byte[] getSelfTestCmd()
Description copied from class: Cmd
Test page print

Specified by:

getSelfTestCmd in class Cmd

Returns:

10.9 public byte[] getBeepCmd()

Description copied from class: Cmd

For Esc Beep

Specified by:

getBeepCmd in class Cmd

Returns:

10.10 public byte[] getAllCutCmd()

Description copied from class: Cmd

For Esc

All cut

Specified by:

getAllCutCmd in class Cmd

Returns:

10.11 public byte[] getDrawLine(int x0,
int y0,
int x1,
int y1,
int lineWidth)

Description copied from class: Cmd

For Tspl, CPCL, ZPL

Draw a line.[only support Horizontal lines and straight lines].

Specified by:

getDrawLine in class Cmd

Parameters:

x0 - start coordinate-x

y0 - start coordinate-y

x1 - end coordinate-x

y1 - end coordinate-y

lineWidth - line width in Dot.

Returns:

10.12 public byte[] getDrawBox(int x0,
int y0,
int x1,
int y1,
int lineWidth)

Description copied from class: Cmd

Draw Box.[For Tspl, CPCL, ZPL]

Specified by:

getDrawBox in class Cmd

Parameters:

x0 - left-top of box coordinate-x
y0 - left-top of box coordinate-y
x1 - right-bottom of box coordinate-x
y1 - right-bottom of box coordinate-y
lineWidth - the line width of box, in dot.

Returns:

10.13 public byte[] getPrinterStatus(PrinterAskStatusEnum printerAskStatusEnum)

Get printer status

Parameters:

printerAskStatusEnum - 1:n = 1: transfer printer status

n = 2: transfer offline status n = 3: transfer error status n = 4: transfer roll sensor

status

Returns:

public byte[] getSetLeftStartSpacing(java.lang.Integer n)

Set the amount of left margin

Parameters:

n - 1mm=n*8

Returns:

10.14 public byte[] getSetAreaWidth(java.lang.Integer n)

Set the print area width

Parameters:

n - 1mm=n*8

Returns:

10.15 public byte[] getReverse(int x0,
int y0,
int x1,
int y1)

Description copied from class: Cmd

For Tspl

Function: Invert the specified area in the image buffer

Specified by:

getReverse in class Cmd

Parameters:

x0 - start coordinate-x

y0 - start coordinate-y

x1 - end coordinate-x

y1 - end coordinate-y

Returns:

10.16 public byte[] getSetXPosition(java.lang.Integer n)

Set absolute print position (x-axis position)

Parameters:

n - Points 1mm=n*8

Returns:

10.17 public byte[] getPageMode(boolean isPageMode)

Set page mode or standard mode. true: page mode, FALSE: standard mode

Parameters:

isPageMode -

Returns:

10.18 public byte[] getPageEnd(boolean isRtnStandard)

Print page schema data

Parameters:

isRtnStandard - true:Back to standard mode, false: Not going back to standard mode

Returns:

10.19 public byte[] getPageArea(int x0,
int y0,
int x1,
int y1)

In page mode, set the print area

Parameters:

x0 - Starting position The position of the x axis

y0 - Starting position Position of the y axis

x1 - End position Position of the X-axis

y1 - End position Position of the Y-axis

Returns:

10.20 public byte[] getTextCmd(TextSetting textSetting,
java.lang.String text)

Print text command

Specified by:

getTextCmd in class Cmd

Parameters:

textSetting - text set

text - charsetname setting, default:GBK

Returns:

11 TsplCmd

11.1getSelfTestCmd

public byte[] getSelfTestCmd()

Test page print

Returns:

11.2 getHeaderCmd

public byte[] getHeaderCmd()

Get header commands

Tspl: CLS command , clear cache

Esc/Pin: printer initialization

Returns:

11.3 getCpclHeaderCmd

```
public byte[] getCpclHeaderCmd(int pageHigh, int pageWidth,int printCopies,int offset)
```

For Cpcl Inital Command

Parameters:

pageHigh - In-mm

pageWidth - In-mm

printCopies - range: 0 ~ n

Returns:

11.4 getPrintCopies

```
public byte[] getPrintCopies(int printCopies)
```

, Print Copies Setting(For Tspl)

Parameters:

printCopies - Default:1

11.5 getEndCmd

```
public byte[] getEndCmd()
```

End command

Returns:

11.6 getLFCmd

```
public byte[] getLFCmd()
```

\n

Wrap

Returns:

11.7 getCRCmd

```
public byte[] getCRCmd()
```

\r

Enter

Returns:

11.8 getLFCRCmd

```
public byte[] getLFCRCmd()
```

\n+\r

Wrap + Enter

Returns:

11.9 getCommonSettingCmd

```
public byte[] getCommonSettingCmd(CommonSetting commonSetting)
```

Common Setting command

Returns:

11.10 getTextCmd

```
public byte[] getTextCmd(TextSetting textSetting, String text, String chartSetName)
```

Print text command

Parameters:

textSetting - , Text Setting

text - , Content

chartSetName - , GBK[Default]、 UTF-8

Returns:

11.11 getTextCmd

```
public byte[] getTextCmd(TextSetting textSetting, String text)
    Print text command
```

Parameters:

textSetting - Text setting
text - chartsetname setting, default:GBK

11.12 getBitmapCmd

```
public byte[] getBitmapCmd(BitmapSetting bitmapSetting, Bitmap bitmap)
    Get Bitmap printing command
```

Parameters:

bitmapSetting - Bitmap setting (width limit)
bitmap - the bitmap which to print

Returns:byte[] cmds

11.13 getBarcodeCmd

```
public byte[] getBarcodeCmd(BarcodeType barcodeType, BarcodeSetting barcodeSetting,
String barcodeContent)
    Get Barcode Printing command
```

Parameters:

barcodeType - Bar code type
barcodeSetting - Bar code setting
barcodeContent - Bar code content

Returns:

11.14 getBeepCmd

```
public byte[] getBeepCmd(int times, int duration)
    For Esc
    Beep
```

Returns:

11.15 getAllCutCmd

```
public byte[] getAllCutCmd()
    For Esc
    All cut
```

Returns:

11.16 getHalfCutCmd

```
public byte[] getHalfCutCmd()
    For Esc
    Half cut
```

Returns:

11.17 getDrawLine

```
public byte[] getDrawLine(int x0,
                           int y0,
                           int x1,
```

```
int y1,  
int lineWidth)
```

For TSpl, CPCL, ZPL

Draw a line.[only support Horizontal lines and straight lines].

Parameters:

x0 - start coordinate-x
y0 - start coordinate-y
x1 - end coordinate-x
y1 - end coordinate-y
lineWidth - line width in Dot.

Returns:

11.18 getDrawBox

```
public byte[] getDrawBox(int x0,  
int y0,  
int x1,  
int y1,  
int lineWidth)
```

Draw Box.[For TSPL, CPCL, ZPL]

Parameters:

x0 - left-top of box coordinate-x
y0 - left-top of box coordinate-y
x1 - right-bottom of box coordinate-x
y1 - right-bottom of box coordinate-y
lineWidth - the line width of box, in dot.

Returns:

11.19 getOpenMoneyBoxCmd

```
public byte[] getOpenMoneyBoxCmd()
```

For Esc

Open Money Box

Returns:

11.20 tscQRCODE

```
public void tscQRCODE(String x,  
String y,  
String eccLevel,  
String cellWidth,  
String rotation,  
String strData)
```

Parameters:

x - X coordinate
y - Y coordinate
eccLevel - Error correction capability level:L=7%,M=15%,Q=25%,H=30%
cellWidth - 1~10
rotation - 0,90,180,270 clockwise
strData - strings, for example: "abc123456"

11.21 getReverse

```
public byte[] getReverse(int x0,  
                        int y0,  
                        int x1,  
                        int y1)
```

For TSPL

Function: Invert the specified area in the image buffer

Parameters:

x0 - start coordinate-x

y0 - start coordinate-y

x1 - end coordinate-x

y1 - end coordinate-y

Returns:

12 CpclCmd

12.1 getSelfTestCmd

```
public byte[] getSelfTestCmd()
```

Test page print

Returns:

12.2 getHeaderCmd

```
public byte[] getHeaderCmd()
```

Get header commands

Tsc: CLS command , clear cache

Esc/Pin: printer initialization

Returns:

12.3 getCpclHeaderCmd

```
public byte[] getCpclHeaderCmd(int pageWidth,  
                               int pageHigh,  
                               int printCopies,  
                               int offset)
```

For Cpcl Initial Command

Parameters:

pageWidth - In-mm

pageHigh - In-mm

printCopies - range: 0 ~ n

Returns:

12.4 getPrintCopies

```
public byte[] getPrintCopies(int printCopies)
```

Parameters:

printCopies - Default:1

12.5 getEndCmd

```
public byte[] getEndCmd()
```

End command

Returns:

12.6 getLFCmd

public byte[] getLFCmd()

\n

Wrap

Returns:

12.7 getCRCmd

public byte[] getCRCmd()

\r

Enter

Returns:

12.8 getLFCRCmd

public byte[] getLFCRCmd()

\n+\r

Wrap + Enter

Returns:

12.9 getCommonSettingCmd

public byte[] getCommonSettingCmd(CommonSetting commonSetting)

Common Setting command

Returns:

12.10 getTextCmd

public byte[] getTextCmd(TextSetting textSetting,

String text,

String chartSetName)

Print text command

Parameters:

textSetting - , Text Setting

text - , Content

chartSetName - , GBK[Default]、UTF-8

Returns:

12.11 getTextCmd

public byte[] getTextCmd(TextSetting textSetting,

String text)

Print text command

Parameters:

textSetting - Text setting

text - charsetname setting, default:GBK

Returns:

12.12 getBitmapCmd

public byte[] getBitmapCmd(BitmapSetting bitmapSetting,

Bitmap bitmap)

Get Bitmap printing command

Parameters:

 bitmapSetting - Bitmap setting (width limit)

 bitmap - the bitmap which to print

Returns: byte[] cmds

12.13 getBarcodeCmd

```
public byte[] getBarcodeCmd(BarcodeType barcodeType,  
                             BarcodeSetting barcodeSetting,  
                             String barcodeContent)
```

 Get Barcode Printing command

Parameters:

 barcodeType - Bar code type

 barcodeSetting - Bar code setting

 barcodeContent - Bar code content

Returns:

12.14 getBeepCmd

```
public byte[] getBeepCmd(int times,  
                           int duration)
```

 For Esc

 Beep

Returns:

12.15 getAllCutCmd

```
public byte[] getAllCutCmd()
```

 For Esc

 All cut

Returns:

12.16 getHalfCutCmd

```
public byte[] getHalfCutCmd()
```

 For Esc

 Half cut

Returns:

12.17 getDrawLine

```
public byte[] getDrawLine(int x0,  
                           int y0,  
                           int x1,  
                           int y1,  
                           int lineWidth)
```

 For TSC, CPCL, ZPL

 Draw a line.[only support Horizontal lines and straight lines].

Parameters:

 x0 - start coordinate-x

 y0 - start coordinate-y

 x1 - end coordinate-x

 y1 - end coordinate-y

 lineWidth - line width in Dot.

Returns:

12.18 getDrawBox

```
public byte[] getDrawBox(int x0,  
                          int y0,  
                          int x1,  
                          int y1,  
                          int lineWidth)
```

Draw Box.[For TSC, CPCL, ZPL]

Parameters:

- x0 - left-top of box coordinate-x
- y0 - left-top of box coordinate-y
- x1 - right-bottom of box coordinate-x
- y1 - right-bottom of box coordinate-y
- lineWidth - the line width of box, in dot.

Returns:

12.19 getOpenMoneyBoxCmd

```
public byte[] getOpenMoneyBoxCmd()
```

For Esc

Open Money Box

Returns:

12.20 CPCL_PrintCGImage

```
public void CPCL_PrintCGImage(String width,  
                               String Height,  
                               String x,  
                               String y,  
                               byte[] data)
```

CPCL CG zac add 20240604

Parameters:

- width -
- Height -
- x -
- y -
- data -

12.21 Cpcl_VBARCODE

```
public void Cpcl_VBARCODE(String barcodetype,  
                           String width,  
                           String ratio,  
                           String height,  
                           String x,  
                           String y,  
                           String data)
```

Cpcl_BARCODE

Parameters:

- barcodetype -

width -
ratio -
height -
x -
y -
data -

12.22 CPCL_QRCODE

```
public void CPCL_QRCODE(String command,  
                        String x,  
                        String y,  
                        String Mn,  
                        String Un,  
                        String data)
```

Print QR code

Parameters:

command - BARCODE VBARCODE
x - :
y -
Mn - QR Code
Un - 1 - 32
data - QR Code H - Q - M - L -

12.23 CPCL_LINE

```
public byte[] CPCL_LINE(int x0,  
                        int y0,  
                        int x1,  
                        int y1,  
                        int width)
```

CPCL drawLine

Parameters:

x0 - X
y0 - Y
x1 -
y1 -
width -

12.24 CPCL_BOX

```
public byte[] CPCL_BOX(int x0,  
                        int y0,  
                        int x1,  
                        int y1,  
                        int width)
```

Parameters:

x0 -
y0 -
x1 -

y1 -
width -

12.25 getReverse

```
public byte[] getReverse(int x0,  
                        int y0,  
                        int x1,  
                        int y1)
```

Function: Invert the specified area in the image buffer

Parameters:

x0 - start coordinate-x
y0 - start coordinate-y
x1 - end coordinate-x
y1 - end coordinate-y

Returns:

13 ZplCmd

13.1 getSelfTestCmd

```
public byte[] getSelfTestCmd()
```

Test page print

Returns:

13.2 getHeaderCmd

```
public byte[] getHeaderCmd()
```

Get header commands

Tspl: CLS command , clear cache

Esc/Pin: printer initialization

Returns:

13.3 getCpclHeaderCmd

```
public byte[] getCpclHeaderCmd(int pageHigh,  
                               int pageWidth,  
                               int printCopies,  
                               int offset)
```

For Cpcl Initial Command

Parameters:

pageHigh - In-mm
pageWidth - In-mm
printCopies - range: 0 ~ n

Returns:

13.4 getPrintCopies

```
public byte[] getPrintCopies(int printCopies)
```

, Print Copies Setting(For Tsc)

Parameters:

printCopies - Default:1

13.5 getEndCmd

public byte[] getEndCmd()

End command

Returns:

13.6 getLFCmd

public byte[] getLFCmd()

\n

Wrap

Returns:

13.7 getCRCmd

public byte[] getCRCmd()

\r

Enter

Returns:

13.8 getLFCRCmd

public byte[] getLFCRCmd()

\n+\r

Wrap + Enter

Returns:

13.9 getCommonSettingCmd

public byte[] getCommonSettingCmd(CommonSetting commonSetting)

Common Setting command

Returns:

13.10 getTextCmd

public byte[] getTextCmd(TextSetting textSetting,
String text,
String chartSetName)
throws UnsupportedOperationException

Print text command

Parameters:

textSetting - , Text Setting

text - , Content

chartSetName - , GBK[Default]、UTF-8

Returns:

13.11 getTextCmd

public byte[] getTextCmd(TextSetting textSetting,
String text)
throws UnsupportedOperationException

Print text command

Parameters:

textSetting - Text setting

text - charsetname setting, default:GBK

Returns:

13.12 getBitmapCmd

```
public byte[] getBitmapCmd(BitmapSetting bitmapSetting,  
                           Bitmap bitmap)
```

Get Bitmap printing command

Parameters:

bitmapSetting - Bitmap setting (width limit)

bitmap - the bitmap which to print

Returns:

byte[] cmds

13.13 getBarcodeCmd

```
public byte[] getBarcodeCmd(BarcodeType barcodeType,  
                           BarcodeSetting barcodeSetting,  
                           String barcodeContent)  
    throws SdkException
```

Get Barcode Printing command

Parameters:

barcodeType - Bar code type

barcodeSetting - Bar code setting

barcodeContent - Bar code content

Returns:

13.14 getBeepCmd

```
public byte[] getBeepCmd(int times,  
                          int duration)
```

For Esc

Beep

Returns:

13.15 getAllCutCmd

```
public byte[] getAllCutCmd()  
    For Esc  
    All cut
```

Returns:

13.16 getHalfCutCmd

```
public byte[] getHalfCutCmd()  
    For Esc  
    Half cut
```

Returns:

13.17 getDrawLine

```
public byte[] getDrawLine(int x0,  
                          int y0,  
                          int x1,  
                          int y1,  
                          int lineWidth)
```

For TSPL, CPCL, ZPL

Draw a line.[only support Horizontal lines and straight lines].

Parameters:

x0 - start coordinate-x
y0 - start coordinate-y
x1 - end coordinate-x
y1 - end coordinate-y
lineWidth - line width in Dot.

Returns:

13.18 getDrawBox

```
public byte[] getDrawBox(int x0,  
                        int y0,  
                        int x1,  
                        int y1,  
                        int lineWidth)
```

Draw Box.[For TSC, CPCL, ZPL]

Parameters:

x0 - left-top of box coordinate-x
y0 - left-top of box coordinate-y
x1 - right-bottom of box coordinate-x
y1 - right-bottom of box coordinate-y
lineWidth - the line width of box, in dot.

Returns:

13.19 getOpenMoneyBoxCmd

```
public byte[] getOpenMoneyBoxCmd()
```

For Esc

13.20 Open Money Box

Returns:

13.21 getZplSumCmds

```
public String getZplSumCmds()  
    private end
```

13.22 getReverse

```
public byte[] getReverse(int x0,  
                        int y0,  
                        int x1,  
                        int y1)
```

For TSC

Function: Invert the specified area in the image buffer

Parameters:

x0 - start coordinate-x
y0 - start coordinate-y
x1 - end coordinate-x
y1 - end coordinate-y

Returns:

14 Printer Example

14.1 Print Self-test page

Example(ESC):

```
CmdFactory cmdFactory = new EscFactory();
Cmd cmd = cmdFactory.create();
cmd.append(cmd.getHeaderCmd());
cmd.append(cmd.getLFCRCmd());
cmd.append(cmd.getSelfTestCmd());
cmd.append(cmd.getLFCRCmd());
rtPrinter.writeMsgAsync(cmd.getAppends());
```

Example(TSPL):

```
CmdFactory cmdFactory = new TsplFactory();
Cmd cmd = cmdFactory.create();
cmd.append(cmd.getHeaderCmd());
cmd.append(cmd.getLFCRCmd());
cmd.append(cmd.getLFCRCmd());
cmd.append(cmd.getSelfTestCmd());
rtPrinter.writeMsgAsync(cmd.getAppends());
```

Example(CPCL):

```
CmdFactory cmdFactory = new CpclFactory();
Cmd cmd = cmdFactory.create();
cmd.append(cmd.getSelfTestCmd());
rtPrinter.writeMsgAsync(cmd.getAppends());
```

Example(ZPL):

```
CmdFactory cmdFactory = new ZplFactory();
Cmd cmd = cmdFactory.create();
cmd.append(cmd.getHeaderCmd());
cmd.append(cmd.getSelfTestCmd());
cmd.append(cmd.getEndCmd());
rtPrinter.writeMsgAsync(cmd.getAppends());
```

14.2 Print text

Example(ESC):

```
private void escPrint() throws UnsupportedOperationException {
    if (rtPrinter != null) {
        EscFactory escFac = new EscFactory();
        EscCmd escCmd = escFac.create();
        escCmd.append(escCmd.getHeaderCmd()); // Initial
        // escCmd.append(((EscCmd)escCmd).getPageMode(true));
        // escCmd.append(((EscCmd)escCmd).getPageArea(10,10,200,200));
    }
}
```

```

//      escCmd.append(((EscCmd)escCmd).getSetLeftStartSpacing(10*8));
//      escCmd.append(((EscCmd)escCmd).getSetAreaWidth(76*8));
//      escCmd.append(((EscCmd)escCmd).getSetAreaWidth(104*8));
escCmd.setChartsetName(mChartsetName);
//      escCmd.append(escCmd.setEscCodePage(255)); // Set codepage
CommonSetting commonSetting = new CommonSetting();
Position txtposition = new Position(0,0);
textSetting.setTxtPrintPosition(txtposition);
commonSetting.setEscLineSpacing(getInputLineSpacing());
escCmd.append(escCmd.getCommonSettingCmd(commonSetting));
escCmd.append(escCmd.getTextCmd(textSetting, printStr));
escCmd.append(escCmd.getLFCRCmd());
escCmd.append(escCmd.getCmdCutNew());
rtPrinter.writeMsgAsync(escCmd.getAppendCmds());
Log.i(TAG, "escPrint: "+ FuncUtils.ByteArrToHex(escCmd.getAppendCmds()));
}
}

```

Example(TSPL):

```

private void tsplPrint() throws UnsupportedOperationException {
    CmdFactory tsplFactory = new TsplFactory();
    Cmd tsplCmd = tsplFactory.create();
    CommonSetting commonSetting = new CommonSetting();
    commonSetting.setLableSizeBean(new LableSizeBean(labelWidth, labelHeight));
    commonSetting.setLabelGap(3);
    commonSetting.setPrintDirection(PrintDirection.NORMAL);
    commonSetting.setSpeedEnum(SpeedEnum.getEnumByString(BaseApplication.labe
elSpeed));

    tsplCmd.append(tsplCmd.getHeaderCmd());
    tsplCmd.append(tsplCmd.getCommonSettingCmd(commonSetting));
    TextSetting textSetting = new TextSetting();
    textSetting.setTsplFontTypeEnum(chooseTscFont);
    int x = 80;
    int y = 80;
    Position position = new Position(x, y);
    textSetting.setTxtPrintPosition(position);
    textSetting.setPrintRotation(PrintRotation.Rotate0);
    textSetting.setxMultiplication(1);
    textSetting.setyMultiplication(1);
    tsplCmd.append(tsplCmd.getTextCmd(textSetting, printStr, mChartsetName));
    try {
        tsplCmd.append(tsplCmd.getPrintCopies(1));
    } catch (SdkException e) {
        e.printStackTrace();
    }
}

```

```

    }
    rtPrinter.writeMsgAsync(tsplCmd.getAppendCmds());
}

```

Example(CPCL):

```

private void cpclPrint() {
    String enlarge = et_enlarge.getText().toString();
    String size = et_cpcl_size.getText().toString();
    String spacing = et_spacing.getText().toString();
    if (TextUtils.isEmpty(size)) {
        size = "0";
    }
    if (TextUtils.isEmpty(enlarge)) {
        enlarge = "1";
    }
    CmdFactory cpclFac = new CpclFactory();
    Cmd cmd = cpclFac.create();
    int printCopies = 1;
    cmd.append(cmd.getCpclHeaderCmd(labelWidth, labelHeight, printCopies,
Integer.parseInt(BaseApplication.labelOffset)); //width, height, copies, offset
    CommonSetting commonSetting = new CommonSetting();
    commonSetting.setSpeedEnum(SpeedEnum.getEnumByString(BaseApplication.labe
elSpeed));
    cmd.append(cmd.getCommonSettingCmd(commonSetting));
    TextSetting textSetting = new TextSetting();
    textSetting.setCpclFontTypeEnum(chooseCpclFont);
    textSetting.setTxtPrintPosition(new Position(0, 0));
    textSetting.setPrintRotation(PrintRotation.Rotate0);
    textSetting.setxMultiplication(Integer.parseInt(enlarge));
    textSetting.setyMultiplication(Integer.parseInt(enlarge));
    textSetting.setCpclFontSize(Integer.parseInt(size));
    textSetting.setBold(ck_bold.isChecked() ? SettingEnum.Enable :
SettingEnum.Disable);
    textSetting.setUnderline(ck_underline.isChecked() ? SettingEnum.Enable :
SettingEnum.Disable);
    int id = rg_align_group.getCheckedRadioButtonId();
    if (id == R.id.rb_align_left) {
        textSetting.setAlign(CommonEnum.ALIGN_LEFT);
    } else if (id == R.id.rb_align_middle) {
        textSetting.setAlign(CommonEnum.ALIGN_MIDDLE);
    } else if (id == R.id.rb_align_right) {
        textSetting.setAlign(CommonEnum.ALIGN_RIGHT);
    }
    if (TextUtils.isEmpty(spacing)) {

```



```

textSetting.setCpclTextSpacing(-1);//-1 = No Setting
} else {
textSetting.setCpclTextSpacing(Integer.parseInt(spacing));
}
try {
cmd.append(cmd.getTextCmd(textSetting, printStr));
} catch (UnsupportedEncodingException e) {
e.printStackTrace();
}
cmd.append(cmd.getEndCmd());
rtPrinter.writeMsgAsync(cmd.getAppendCmds());
}

```

Example(ZPL):

```

private void zplTextPrint() {
try {
CmdFactory zplFac = new ZplFactory();
Cmd zplCmd = zplFac.create();
zplCmd.append(zplCmd.getHeaderCmd());
CommonSetting commonSetting = new CommonSetting();
commonSetting.setLableSizeBean(new LableSizeBean(labelWidth, labelHeight));
commonSetting.setPrintDirection(PrintDirection.NORMAL);
zplCmd.append(zplCmd.getCommonSettingCmd(commonSetting));
TextSetting textSetting = new TextSetting();
textSetting.setZplFontTypeEnum(ZplFontTypeEnum.FONT_DOWNLOAD_FONT);
textSetting.setZplHeightFactor(30);// >10
textSetting.setZplWidthFactor(30);// >10
textSetting.setTxtPrintPosition(new Position(80, 80));
textSetting.setPrintRotation(PrintRotation.Rotate0);
zplCmd.append(zplCmd.getTextCmd(textSetting, printStr, mChartsetName));
zplCmd.append(zplCmd.getPrintCopies(1));
zplCmd.append(zplCmd.getEndCmd());
if (rtPrinter != null) {
rtPrinter.writeMsgAsync(zplCmd.getAppendCmds());
}
} catch (Exception e) {
e.printStackTrace();
}
}
}

```

14.3 Print Barcode

Example(ESC):

```

private void escPrint(){

```

```

        try {
            CmdFactory cmdFactory = new EscFactory();
            Cmd escCmd = cmdFactory.create();
            escCmd.append(escCmd.getHeaderCmd());
            CommonSetting commonSetting = new CommonSetting();
            int commonSettingAlign = commonSetting.getAlign();
            commonSetting.setAlign(CommonEnum.ALIGN_MIDDLE);//Centering instruction
            escCmd.append(escCmd.getCommonSettingCmd(commonSetting));
            BarcodeSetting barcodeSetting = new BarcodeSetting();
            barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)
;

            barcodeSetting.setHeightInDot(72);//accept value:1~255
            barcodeSetting.setBarcodeWidth(3);//accept value:2~6
            //          barcodeSetting.setQrcodeDotSize(15);//accept value: Esc(1~25),
Tsc(1~10)

            barcodeSetting.setEscBarcodeFont(ESCBarcodeFontTypeEnum.BARFONT_A_12x24);
            escCmd.append(escCmd.getBarcodeCmd(barcodeType, barcodeSetting,
barcodeContent));//Barcode type, barcode settings, barcode content
            escCmd.append(escCmd.getLFCRCmd());
            escCmd.append(escCmd.getCmdCutNew());
            commonSetting.setAlign(commonSettingAlign);//Set it back after using it.
            escCmd.append(escCmd.getCommonSettingCmd(commonSetting));
            rtPrinter.writeMsgAsync(escCmd.getAppendCmds());
        } catch (SdkException e) {
            e.printStackTrace();
        }
    }
}

```

Example(TSPL):

```

private void tscPrint() {
    try {
        CmdFactory tscFac = new TsplFactory();
        Cmd tscCmd = tscFac.create();
        tscCmd.append(tscCmd.getHeaderCmd());
        CommonSetting commonSetting = new CommonSetting();
        commonSetting.setLableSizeBean(new LableSizeBean(labelWidth, labelHeight));
        commonSetting.setLabelGap(3);
        commonSetting.setPrintDirection(PrintDirection.NORMAL);
        tscCmd.append(tscCmd.getCommonSettingCmd(commonSetting));
        BarcodeSetting barcodeSetting = new BarcodeSetting();
        barcodeSetting.setNarrowInDot(2);//narrow bar setting, bar width
        barcodeSetting.setWideInDot(4);
        barcodeSetting.setHeightInDot(48);//bar height setting
        barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)

```

```

;
        barcodeSetting.setPrintRotation(printRotation);
        barcodeSetting.setPosition(new Position(30, 80));
        byte[] barcodeCmd = tscCmd.getBarcodeCmd(barcodeType, barcodeSetting,
barcodeContent);
        tscCmd.append(barcodeCmd);
        tscCmd.append(tscCmd.getPrintCopies(1));
        tscCmd.append(tscCmd.getEndCmd());
        if (rtPrinter != null) {
            rtPrinter.writeMsgAsync(tscCmd.getAppendCmds());
        }
    } catch (SdkException e) {
        throw new RuntimeException(e);
    }
}

```

Example(CPCL):

```

private void cpclPrint() {
    try {
        CmdFactory cpclFac = new CpclFactory();
        Cmd cmd = cpclFac.create();
        cmd.append(cmd.getCpclHeaderCmd(labelWidth, labelHeight, 1,
Integer.parseInt(BaseApplication.labelOffset)));
        BarcodeSetting barcodeSetting = new BarcodeSetting();
        barcodeSetting.setPrintRotation(printRotation);
        barcodeSetting.setNarrowInDot(0);//narrow bar width
        barcodeSetting.setWideInDot(0);
        barcodeSetting.setBarcodeStringPosition(ck_show_text.isChecked() ?
BarcodeStringPosition.BELOW_BARCODE : BarcodeStringPosition.NONE);
        if (printRotation == PrintRotation.Rotate0) {
            barcodeSetting.setPosition(new Position(10, 20));//bar height setting
        } else {
            barcodeSetting.setPosition(new Position(300, 300));//bar height setting
        }
        barcodeSetting.setHeightInDot(56);
        byte[] barcodeCmd = new byte[0];
        barcodeCmd = cmd.getBarcodeCmd(barcodeType, barcodeSetting,
barcodeContent);
        cmd.append(barcodeCmd);
        cmd.append(cmd.getEndCmd());
        if (rtPrinter != null) {
            rtPrinter.writeMsgAsync(cmd.getAppendCmds());
        }
    } catch (SdkException e) {

```

```
throw new RuntimeException(e);  
}  
}
```

Example(ZPL):

```
private void zplPrint() {  
    try {  
        CmdFactory zplFac = new ZplFactory();  
        Cmd zplCmd = zplFac.create();  
        zplCmd.append(zplCmd.getHeaderCmd());  
        CommonSetting commonSetting = new CommonSetting();  
        commonSetting.setLableSizeBean(new LableSizeBean(labelWidth, labelHeight));  
        commonSetting.setLabelGap(2);  
        commonSetting.setPrintDirection(PrintDirection.NORMAL);  
        zplCmd.append(zplCmd.getCommonSettingCmd(commonSetting));  
        BarcodeSetting barcodeSetting = new BarcodeSetting();  
        barcodeSetting.setHeightInDot(48);  
        barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)  
;  
  
        barcodeSetting.setPrintRotation(printRotation);  
        int x = 10, y = 80;  
        switch (printRotation) {  
            case Rotate0:  
                x = 10;  
                y = 10;  
                break;  
            case Rotate90:  
                x = (labelWidth * 8) / 2;  
                y = 20;  
                break;  
            case Rotate270:  
                x = (labelWidth * 8) / 2;  
                y = (labelHeight * 8) - 20;  
                break;  
            default:  
                break;  
        }  
        barcodeSetting.setPosition(new Position(x, y));  
        byte[] barcodeCmd = zplCmd.getBarcodeCmd(barcodeType, barcodeSetting,  
barcodeContent);  
        zplCmd.append(barcodeCmd);  
        zplCmd.append(zplCmd.getPrintCopies(1));  
        zplCmd.append(zplCmd.getEndCmd());  
    }  
}
```

```

if (rtPrinter != null) {
    rtPrinter.writeMsgAsync(zplCmd.getAppendCmds());
}
} catch (SdkException e) {
    throw new RuntimeException(e);
}
}

```

14.4 Print Bitmap

Example(ESC):

```

CmdFactory cmdFactory = new EscFactory();
Cmd cmd = cmdFactory.create();
cmd.append(cmd.getHeaderCmd());
CommonSetting commonSetting = new CommonSetting();
commonSetting.setAlign(CommonEnum.ALIGN_MIDDLE);
cmd.append(cmd.getCommonSettingCmd(commonSetting));
BitmapSetting bitmapSetting = new BitmapSetting();
bitmapSetting.setBmpPrintMode(BmpPrintMode.MODE_SINGLE_FAST);
bitmapSetting.setBimtapLimitWidth(bmpPrintWidth* 8);
try {
    cmd.append(cmd.getBitmapCmd(bitmapSetting, mBitmap));
} catch (SdkException e) {
    e.printStackTrace();
}
cmd.append(cmd.getLFCRCmd());
cmd.append(cmd.getCmdCutNew());
rtPrinter.writeMsg(cmd.getAppendCmds());

```

Example(TSPL):

```

private void tsplPrint(){
    new Thread(new Runnable() {
        @Override
        public void run() {
            showProgressDialog(null);
            CmdFactory tsplFactory = new TsplFactory();
            Cmd tsplCmd = tsplFactory.create();
            tsplCmd.append(tsplCmd.getHeaderCmd());
            CommonSetting commonSetting = new CommonSetting();
            commonSetting.setLableSizeBean(new LableSizeBean(bmpPrintWidth,
bmpPrintHeight));
            commonSetting.setLabelGap(-1);
            commonSetting.setPrintDirection(PrintDirection.NORMAL);
            commonSetting.setLabelCutEnum(getCuttype());//set cut
Disable,CUTTER,PEEL,TEAR

```

```

tsplCmd.append(tsplCmd.getHeaderCmd());
tsplCmd.append(tsplCmd.getCommonSettingCmd(commonSetting));
BitmapSetting bitmapSetting = new BitmapSetting();
bitmapSetting.setPrintPostion(new Position(0, 0));
bitmapSetting.setBimtapLimitWidth(bmpPrintWidth * 8);
bitmapSetting.setBmpPrintMode(BmpPrintMode.MODE_SINGLE_COLOR);
try {
tsplCmd.append(tsplCmd.getBitmapCmd(bitmapSetting, mBitmap));
tsplCmd.append(tsplCmd.getPrintCopies(1)); //Print Copies setting
} catch (SdkException e) {
e.printStackTrace();
}
if (rtPrinter != null) {
rtPrinter.writeMsg(tsplCmd.getAppndCmds());
}
hideProgressDialog();
}
}).start();
}

```

Example(CPCL):

```

private void cpclPrint() {
new Thread(new Runnable() {
@Override
public void run() {
try {
showProgressDialog(null);
CmdFactory cpclFactory = new CpclFactory();
Cmd cmd = cpclFactory.create();
cmd.append(cmd.getCpclHeaderCmd(bmpPrintWidth, bmpPrintHeight, 1, 0));
BitmapSetting bitmapSetting = new BitmapSetting();
bitmapSetting.setPrintPostion(new Position(0, 0));
bitmapSetting.setBimtapLimitWidth(bmpPrintWidth * 8);
bitmapSetting.setBimtapLimitHeight(bmpPrintHeight * 8);
bitmapSetting.setBmpPrintMode(BmpPrintMode.MODE_MULTI_COLOR);
cmd.append(cmd.getBitmapCmd(bitmapSetting, mBitmap));
cmd.append(cmd.getEndCmd());
if (rtPrinter != null) {
rtPrinter.writeMsg(cmd.getAppndCmds());
}
} catch (SdkException e) {
e.printStackTrace();
}
hideProgressDialog();
}
}
}

```

```
}  
}).start();  
}
```

Example(ZPL):

```
private void zplPrint() {  
    new Thread(new Runnable() {  
        @Override  
        public void run() {  
            showProgressDialog(null);  
            CmdFactory zplFac = new ZplFactory();  
            Cmd zplCmd = zplFac.create();  
            zplCmd.append(zplCmd.getHeaderCmd());  
            CommonSetting commonSetting = new CommonSetting();  
            commonSetting.setLableSizeBean(new LableSizeBean bmpPrintWidth,  
bmpPrintHeight));  
            //                commonSetting.setLabelGap(3);  
            commonSetting.setPrintDirection(PrintDirection.NORMAL);  
            zplCmd.append(zplCmd.getHeaderCmd());  
            zplCmd.append(zplCmd.getCommonSettingCmd(commonSetting));  
            BitmapSetting bitmapSetting = new BitmapSetting();  
            bitmapSetting.setPrintPostion(new Position(0, 0));  
            bitmapSetting.setBimtapLimitWidth(bmpPrintWidth * 8);  
            try {  
                zplCmd.append(zplCmd.getBitmapCmd(bitmapSetting, mBitmap));  
                zplCmd.append(zplCmd.getPrintCopies(1));  
            } catch (SdkException e) {  
                e.printStackTrace();  
            }  
            zplCmd.append(zplCmd.getEndCmd());  
            if (rtPrinter != null) {  
                rtPrinter.writeMsg(zplCmd.getAppndCmds());  
            }  
            hideProgressDialog();  
        }  
    }).start();  
}
```

14.5 Print Template

Example(ESC):

```
public void escTemplet() throws UnsupportedEncodingException, SdkException {  
    CmdFactory escFac = new EscFactory();  
    Cmd escCmd = escFac.create();
```

```

        escCmd.setCharset("UTF-8");
        TextSetting textSetting = new TextSetting();
        textSetting.setAlign(CommonEnum.ALIGN_MIDDLE);
        textSetting.setBold(SettingEnum.Enable);
        textSetting.setUnderline(SettingEnum.Disable);
        textSetting.setIsAntiWhite(SettingEnum.Disable);
        textSetting.setDoubleHeight(SettingEnum.Enable);
        textSetting.setDoubleWidth(SettingEnum.Enable);
        textSetting.setItalic(SettingEnum.Disable);
        textSetting.setIsEscSmallCharactor(SettingEnum.Disable);
        escCmd.append(escCmd.getHeaderCmd());
        escCmd.append(escCmd.getTextCmd(textSetting, title));
        escCmd.append(escCmd.getLFCRCmd());
        textSetting.setIsEscSmallCharactor(SettingEnum.Enable);
        textSetting.setBold(SettingEnum.Disable);
        textSetting.setDoubleHeight(SettingEnum.Disable);
        textSetting.setDoubleWidth(SettingEnum.Disable);
        escCmd.append(escCmd.getTextCmd(textSetting, content_tel));
        escCmd.append(escCmd.getLFCRCmd());
        textSetting.setUnderline(SettingEnum.Enable);
        escCmd.append(escCmd.getTextCmd(textSetting, content_email));
        escCmd.append(escCmd.getLFCRCmd());
        escCmd.append(escCmd.getLFCRCmd());
        BitmapSetting bitmapSetting = new BitmapSetting();
        bitmapSetting.setBimtapLimitWidth(40);
        Bitmap bmp = BitmapFactory.decodeResource(context.getResources(),
R.mipmap.ic_launcher);
        escCmd.append(escCmd.getBitmapCmd(bitmapSetting,
Bitmap.createBitmap(bmp)));
        escCmd.append(escCmd.getLFCRCmd());
        BarcodeSetting barcodeSetting = new BarcodeSetting();
        barcodeSetting.setEscBarcodePrintOritention(EscBarcodePrintOritention.Rotate0);
        barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)
;

        barcodeSetting.setHeightInDot(72);//accept value:1~255
        barcodeSetting.setBarcodeWidth(3);//accept value:2~6
        escCmd.append(escCmd.getBarcodeCmd(BarcodeType.CODE128, barcodeSetting,
barcode_str));
        escCmd.append(escCmd.getLFCRCmd());
        barcodeSetting.setQrcodeDotSize(5);//accept value: Esc(1~15), Tsc(1~10)
        barcodeSetting.setQrcodeEccLevel(QrcodeEccLevel.L);
        escCmd.append(escCmd.getBarcodeCmd(BarcodeType.QR_CODE, barcodeSetting,
content_email));

```



```
//      escCmd.append(escCmd.getLFCRCmd());
escCmd.append(escCmd.getLFCRCmd());
escCmd.append(escCmd.getCmdCutNew());
rtPrinter.writeMsgAsync(escCmd.getAppenCmds());
}
```

Example(TSPL):

```
public void tsplPrint() throws SdkException {
    CmdFactory tsplFactory = new TsplFactory();
    Cmd cmd = tsplFactory.create();
    cmd.append(cmd.getHeaderCmd());
    CommonSetting commonSetting = new CommonSetting();
    commonSetting.setLabelGap(3);
    commonSetting.setPrintDirection(PrintDirection.NORMAL);
    commonSetting.setLableSizeBean(new LableSizeBean(80, 80)); //label width =
80mm, label height = 80mm
    cmd.append(cmd.getCommonSettingCmd(commonSetting));
    //(int x0, int y0, int x1, int y1, int lineWidth)
    cmd.append(cmd.getDrawLine(50,50,180,50,2));
    cmd.append(cmd.getDrawLine(50,100,50,200,2));
    TextSetting textSetting = new TextSetting();
    textSetting.setTsplFontTypeEnum(TsplFontTypeEnum.Font_TSS24_BF2_For_Simple
_Chinese);
    textSetting.setTxtPrintPosition(new Position(80, 80));
    textSetting.setPrintRotation(PrintRotation.Rotate0);
    textSetting.setxMultiplication(1);
    textSetting.setyMultiplication(1);
    try {
        cmd.append(cmd.getTextCmd(textSetting, title));
        textSetting.setTxtPrintPosition(new Position(120, 140));
        textSetting.setxMultiplication(1);
        textSetting.setyMultiplication(1);
        textSetting.setTsplFontTypeEnum(TsplFontTypeEnum.Font_TSS24_BF2_For_Simple
_Chinese);
        cmd.append(cmd.getTextCmd(textSetting, content_tel));
        textSetting.setTxtPrintPosition(new Position(80, 180));
        cmd.append(cmd.getTextCmd(textSetting, content_email));
    } catch (UnsupportedEncodingException e) {
        e.printStackTrace();
    }
    BitmapSetting bitmapSetting = new BitmapSetting();
    bitmapSetting.setBimtapLimitWidth(40);
    bitmapSetting.setPrintPostion(new Position(250, 220));
    Bitmap bmp = BitmapFactory.decodeResource(context.getResources(),
```

```

R.mipmap.ic_launcher);
    cmd.append(cmd.getBitmapCmd(bitmapSetting, Bitmap.createBitmap bmp));
    cmd.append(cmd.getLFCRCmd());
    BarcodeSetting barcodeSetting = new BarcodeSetting();
    barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)
;

    barcodeSetting.setPosition(new Position(180, 280));
    barcodeSetting.setPrintRotation(PrintRotation.Rotate0);
    barcodeSetting.setHeightInDot(72);//accept value:1~255
    barcodeSetting.setBarcodeWidth(3);//accept value:2~6
    cmd.append(cmd.getBarcodeCmd(BarcodeType.CODE128, barcodeSetting,
barcode_str));

    cmd.append(cmd.getLFCRCmd());

    barcodeSetting.setPosition(new Position(220, 420));
    barcodeSetting.setQrcodeDotSize(5);//accept value: Esc(1~15), Tsc(1~10)
    barcodeSetting.setQrcodeEccLevel(QrcodeEccLevel.L);
    cmd.append(cmd.getBarcodeCmd(BarcodeType.QR_CODE, barcodeSetting,
content_email));
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getPrintCopies(1));
    cmd.append(cmd.getEndCmd());
    rtPrinter.writeMsgAsync(cmd.getAppends());

}

```

Example(CPCL):

```

public void cpclPrint() throws UnsupportedOperationException, SdkException {
    CmdFactory cpclFac = new CpclFactory();
    Cmd cmd = cpclFac.create();
    cmd.append(cmd.getCpclHeaderCmd(80, 80, 1, 0));
    TextSetting textSetting = new TextSetting();
    textSetting.setCpclFontTypeEnum(CpclFontTypeEnum.Font_3);
    textSetting.setTxtPrintPosition(new Position(80, 80));
    textSetting.setPrintRotation(PrintRotation.Rotate0);
    textSetting.setxMultiplication(2);
    textSetting.setyMultiplication(2);
    try {
        cmd.append(cmd.getTextCmd(textSetting, title));
        textSetting.setTxtPrintPosition(new Position(120, 140));
        textSetting.setxMultiplication(1);
    }
}

```

```

        textSetting.setyMultiplication(1);
        textSetting.setCpclFontTypeEnum(CpclFontTypeEnum.Font_2);
        cmd.append(cmd.getTextCmd(textSetting, content_tel));
        textSetting.setTxtPrintPosition(new Position(80, 180));
        cmd.append(cmd.getTextCmd(textSetting, content_email));
    } catch (UnsupportedEncodingException e) {
        e.printStackTrace();
    }

    BitmapSetting bitmapSetting = new BitmapSetting();
    bitmapSetting.setBimtapLimitWidth(40);
    bitmapSetting.setPrintPostion(new Position(250, 220));

    Bitmap bmp = BitmapFactory.decodeResource(context.getResources(),
R.mipmap.ic_launcher);
    cmd.append(cmd.getBitmapCmd(bitmapSetting, Bitmap.createBitmap(bmp)));
    cmd.append(cmd.getLFCRCmd());

    BarcodeSetting barcodeSetting = new BarcodeSetting();
barcodeSetting.setEscBarcodePrintOritention(EscBarcodePrintOritention.Rotate0);
barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE);
    barcodeSetting.setPosition(new Position(180, 280));
    barcodeSetting.setPrintRotation(PrintRotation.Rotate0);
    barcodeSetting.setHeightInDot(72);//accept value:1~255
    barcodeSetting.setBarcodeWidth(3);//accept value:2~6
    cmd.append(cmd.getBarcodeCmd(BarcodeType.CODE128, barcodeSetting,
barcode_str));
    cmd.append(cmd.getLFCRCmd());
    barcodeSetting.setPosition(new Position(220, 420));
    barcodeSetting.setQrcodeDotSize(5);//accept value: Esc(1~15), Tsc(1~10)
    barcodeSetting.setQrcodeEccLevel(QrcodeEccLevel.L);
    cmd.append(cmd.getBarcodeCmd(BarcodeType.QR_CODE, barcodeSetting,
content_email));
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getLFCRCmd());
    cmd.append(cmd.getEndCmd());
    rtPrinter.writeMsgAsync(cmd.getAppendCmds());
}

```

Example(ZPL):

```

public void zplPrint() throws UnsupportedEncodingException, SdkException {
    CmdFactory fac = new ZplFactory();
    Cmd cmd = fac.create();
    cmd.append(cmd.getHeaderCmd());
}

```

```

CommonSetting commonSetting = new CommonSetting();
commonSetting.setLabelGap(2);
commonSetting.setPrintDirection(PrintDirection.REVERSE);
commonSetting.setLableSizeBean(new LableSizeBean(80, 80)); //label width =
80mm, label height = 80mm
cmd.append(cmd.getCommonSettingCmd(commonSetting));
TextSetting textSetting = new TextSetting();
textSetting.setZplFontTypeEnum(ZplFontTypeEnum.FONT_DOWNLOAD_FONT);
textSetting.setTxtPrintPosition(new Position(130, 80));
textSetting.setPrintRotation(PrintRotation.Rotate0);
textSetting.setZplHeightFactor(64); // >10
textSetting.setZplWidthFactor(64); // >10
try {
cmd.append(cmd.getTextCmd(textSetting, title));
textSetting.setTxtPrintPosition(new Position(80, 160));
textSetting.setZplHeightFactor(2); // 1~10
textSetting.setZplWidthFactor(2); // 1~10
textSetting.setZplFontTypeEnum(ZplFontTypeEnum.FONT_2);
cmd.append(cmd.getTextCmd(textSetting, content_tel));
textSetting.setTxtPrintPosition(new Position(80, 180));
cmd.append(cmd.getTextCmd(textSetting, content_email));
} catch (UnsupportedEncodingException e) {
e.printStackTrace();
}
BitmapSetting bitmapSetting = new BitmapSetting();
bitmapSetting.setBimtapLimitWidth(40);
bitmapSetting.setPrintPostion(new Position(250, 220));
Bitmap bmp = BitmapFactory.decodeResource(context.getResources(),
R.mipmap.ic_launcher);
try {
cmd.append(cmd.getBitmapCmd(bitmapSetting, Bitmap.createBitmap(bmp)));
} catch (SdkException e) {
e.printStackTrace();
}
cmd.append(cmd.getLFCRCmd());
BarcodeSetting barcodeSetting = new BarcodeSetting();
barcodeSetting.setBarcodeStringPosition(BarcodeStringPosition.BELOW_BARCODE)
;

barcodeSetting.setPosition(new Position(120, 280));
barcodeSetting.setPrintRotation(PrintRotation.Rotate0);
barcodeSetting.setHeightInDot(72); //accept value: 1~255
barcodeSetting.setBarcodeWidth(3); //accept value: 2~6
try {
cmd.append(cmd.getBarcodeCmd(BarcodeType.CODE128, barcodeSetting,

```

```

barcode_str));
        } catch (SdkException e) {
            e.printStackTrace();
        }
        cmd.append(cmd.getLFCRCmd());
        barcodeSetting.setPosition(new Position(200, 420));
        barcodeSetting.setQrcodeDotSize(5);//accept value: Esc(1~15), Tsc(1~10)
        barcodeSetting.setQrcodeEccLevel(QrcodeEccLevel.L);
        try {
            cmd.append(cmd.getBarcodeCmd(BarcodeType.QR_CODE, barcodeSetting,
content_email));
        } catch (SdkException e) {
            e.printStackTrace();
        }
        cmd.append(cmd.getLFCRCmd());
        cmd.append(cmd.getLFCRCmd());
        cmd.append(cmd.getLFCRCmd());
        try {
            cmd.append(cmd.getPrintCopies(1));//ZPL must add this function, print copies
settings,
        } catch (SdkException e) {
            e.printStackTrace();
        }
        cmd.append(cmd.getEndCmd());
        String str = new String(cmd.getAppendCmds());
        Log.e("sss", str);
        rtPrinter.writeMsg(cmd.getAppendCmds());
    }

```

14.6 getPrintStauts

The message sent by the printer is retrieved in the monitor "printerReadMsgCallback(PrinterInterface printerInterface, final byte[] bytes)".

```

private void getPrintStauts() {
    msgType = 1;
    CmdFactory cmdFactory = new EscFactory();
    Cmd cmd = cmdFactory.create();
    rtPrinter.writeMsgAsync(cmd.getPrintStausCmd PrintStatusCmd.cmd_Normal));
}

```

14.7 getFirmwareVersion

The message sent by the printer is retrieved in the monitor "printerReadMsgCallback(PrinterInterface printerInterface, final byte[] bytes)".

```

private void getFirmwareVersion() {
    msgType = 3;
    if (BaseApplication.getInstance().getCurrentCmdType() != BaseEnum.CMD_ESC) {

```

```
rtPrinter.writeMsgAsync(new TsplFactory().create().getLabelFirmwareversion());  
} else {  
rtPrinter.writeMsgAsync(new EscFactory().create().getEscFirmwareversion());  
}  
}
```